

Department of Information Technology
Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch : IV Semester-B. Tech Information Technology
Course Code & Title : IT3401 Web Essentials
Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 3 /Built-in Objects
- **Course Outcome** : CO3
- **Topic Learning Outcome:** TLO25
- **Activity Chosen** : Reflection
- **Justification:**
 - Reflection activity helps students evaluate their understanding of concepts and identify areas where they need additional clarification.
 - JavaScript Built-in Objects play a vital role in web development by providing predefined functionalities for handling strings, numbers, dates, arrays, and mathematical operations efficiently.
 - This activity was chosen to help students understand the practical use of JavaScript Built-in Objects and how they simplify programming tasks in real-time web application development.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - The activity was conducted at the end of the class after a detailed explanation of JavaScript Built-in Objects, including String, Array, Math, Date, and Number objects with suitable examples and demonstrations, as shown in Figure 1.
 - The instructor explained how built-in objects simplify programming tasks, minimize coding effort and improve the efficiency of JavaScript programs.
 - Students reflected on the concepts learned during the session and identified difficulties in understanding the usage of String and Array methods in different programming scenarios.
 - After collecting the responses, the instructor analyzed their feedback and addressed the doubts in the following class, as shown in Figure 2.
 - This activity helped students strengthen their conceptual understanding and improve their confidence in applying JavaScript Built-in Objects in programming tasks.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO10	PSO1
CO3	2	2	2	2	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO5	PO10	PSO1
	2	2	2	2	1	3
Justification for correlation	Students will be able to understand the purpose and functionality of JavaScript Built-in Objects in web programming.	Students will be able to analyze programming problems and identify appropriate built-in methods for solving them efficiently.	Students will be able to develop JavaScript programs using built-in objects to handle real-world data processing tasks.	Students will be able to utilize JavaScript Built-in Objects and browser-based programming tools effectively during application development.	Students will be able to communicate their understanding of JavaScript concepts and clarify programming doubts through reflection and discussion.	Students will be able to apply JavaScript Built-in Objects in designing interactive and efficient web applications.

- **Images / Screenshot of the practice:**



Fig 1: Reflection Activity

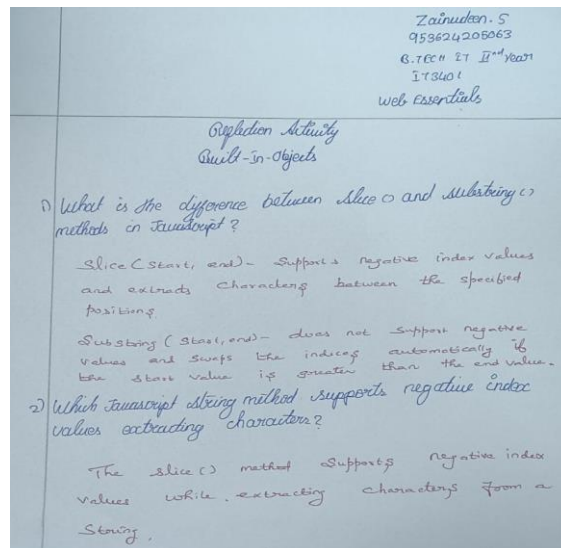


Fig 2: Sample Sheet of Reflection Activity

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students were encouraged to self-evaluate their understanding of JavaScript Built-in Objects and identify methods that required further clarification.
- Students expressed that the reflection activity provided them with an opportunity to share their doubts openly and interact more effectively with the instructor.

❖ Benefit of the practice:

- The activity encouraged students to think critically about the functionality and usage of different JavaScript Built-in methods.
- The activity inspired students to self-assess their grasp of JavaScript Built-in Objects and determine the concepts that needed more attention and reinforcement.

❖ Challenges faced in implementation:

- Some students found it difficult to differentiate between similar methods of String and Array objects.
- Few students hesitated to share their learning difficulties openly in the reflection activity, creating challenges in assessing individual conceptual understanding.

In the next class, the instructor resolved the doubts associated with JavaScript Built-in Objects by providing additional coding examples and real-time programming scenarios.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/RE_C_BV_22-23_U%20III.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1.KVS_Reflection_FDS.pdf

Signature of Faculty Member

HOD